Gargoyle Gecko Care Guide

Rhacodactylus auriculatus

Provide broad stable climbing surfaces, dense cover, a complete gecko diet and regular insect meals. Individual housing limits tail injuries, food competition and unplanned reproduction.

Species Profile

Common name: Gargoyle Gecko

Scientific name: *Rhacodactylus auriculatus*

Adult size: 8-10 in (20-25 cm) total length

Average lifespan: 15-20 years

Native range: Grande Terre, New Caledonia

Habitat and activity: Mostly nocturnal; shrubs, low trees and forest/maquis edges

Diet: Complete crested/gargoyle-gecko diet and regular varied insects

Care difficulty: Beginner to Intermediate

Shopping List

- Secure enclosure at least 18 × 18 × 36 in (46 × 46 × 91 cm; L × W × H) for one adult; see Housing for exceptions and group arrangements.
- Digital thermometers for different zones; a suitable hygrometer; timer for a consistent day/night cycle.
- Species-appropriate substrate and multiple stable hides or covered retreats.
- Reliable supply of the staple foods described below, clean feeding tools and a transport container.
- Appropriate overhead heat source if required, controlled by a compatible thermostat; infrared thermometer for surfaces.
- Appropriate visible lighting and, where recommended below, a linear UVB fixture with measured shade and exposure zones.
- Fresh-water dish or cup; insect supplements when applicable; a scale appropriate to the animal's size.

Housing

Our recommended starting adult enclosure is 18 × 18 × 36 in (46 × 46 × 91 cm; L × W × H) for one animal. Larger usable space is encouraged. These dimensions are a husbandry recommendation, not a species-specific experimentally proven threshold.

Fill the height with firmly secured horizontal and diagonal branches, cork, foliage and sheltered paths. Provide several resting places at different heights and enough cover to travel between environmental zones without crossing a bare open box.

Juveniles may use a furnished adult enclosure when food intake, hydration and waste can be monitored. A temporary grow-out enclosure should still offer choice and cover. Quarantine newcomers separately with dedicated tools and a substrate that permits health monitoring; arrange veterinary assessment for unknown-origin animals.

Lighting & UVB

Provide roughly 10-12 hours of visible daylight followed by darkness. A measured UV Index (UVI) around 0.5-1.0 at the most exposed usable site, declining to near zero inside retreats, is our conservative starting recommendation. Low-level UVB is recommended when a usable gradient can be provided; nocturnal activity alone is not a reason to exclude light.

Measure exposure at the animal's back height with an appropriate UVI meter, such as a Solarmeter 6.5. Lamp percentage, wattage and a fixed distance alone do not establish exposure: reflectors, mesh, perch height, plant growth and lamp age change output. Keep glass/plastic out of the UV path, follow fixture safety instructions and provide complete shade. Replace or reassess lamps according to measured output and manufacturer guidance.

Heating & Temperature

- Localized basking surface: 82-84°F (28-29°C).
- Cool-zone air: 70-75°F (21-24°C).
- Nighttime air: 65-72°F (18-22°C) for healthy established animals.

Create the gradient on one side rather than heating every hide to the same temperature. Use a compatible thermostat on each heat source; a manual lamp dimmer does not replace thermostatic protection. Secure the probe and verify temperatures independently: probe thermometers measure air/retreat temperature, while an infrared thermometer measures exposed surfaces.

Use appropriately guarded overhead warmth when needed. Keep animals clear of hot fixtures, avoid heat rocks, and switch visible lights off at night. Night heat is needed only when the room falls below the appropriate range; use a thermostat-controlled lightless source if necessary.

Do not allow the whole enclosure to approach the warm-site temperature. Sustained mid-80s°F (about 29°C) air is a warning for these setups; heat-sensitive individuals may need intervention sooner. Keep a cooler refuge available and respond to prolonged heat avoidance.

Humidity & Ventilation

50-70% during drying; brief rises to 80-90% after misting. Treat readings as a starting pattern rather than a number to chase regardless of substrate, airflow or animal behavior.

Measure humidity in a representative occupied zone, away from a freshly sprayed surface. Provide sheltered moisture choices while retaining airflow. Persistent condensation, sour odor, flooded soil or surfaces that never dry indicate a setup problem. Ventilation and hydration must be adjusted together rather than simply sealing all vents.

Water & Hydration

Provide a clean, appropriately shallow dish or accessible elevated water cup at all times. Change it daily and immediately when soiled; wash it regularly with dedicated tools. Potable water is generally suitable, with treatment where local water quality requires it.

For animals that drink droplets, mist foliage or suitable surfaces at the time they become active, then allow partial drying. Droplets supplement the dish; water chemistry claims should be based on evidence rather than a species' trade name.

Substrate

For established juveniles and adults, use 2-4 inches of a moisture-retentive tropical substrate such as a pesticide- and fertilizer-free organic topsoil/coco mix, ReptiSoil-style blend, or ABG-type mix. Add clean leaf litter over the surface. Bioactive enclosures should include adequate drainage, a screen separator, deeper substrate, live plants, and a cleanup crew appropriate for the setup.

Paper towel is appropriate for quarantine, medical monitoring, or very small juveniles while feeding and stool output are being established. Avoid reptile carpet, pine or cedar, large bark chips, and other coarse particles that can catch toes or be difficult to pass if swallowed. Spot-clean waste and spoiled food promptly, and refresh or replace non-bioactive substrate as it becomes soiled rather than waiting for odor or mold problems.

Food & Feeding

Gargoyle geckos are omnivores, and field studies show a higher proportion of animal prey than is typical for many other New Caledonian geckos. In captivity, use a reputable nutritionally complete crested/gargoyle gecko diet as the staple and provide live insects regularly for protein, hunting behavior, and dietary variety. Complete diets should be mixed according to the manufacturer's directions and replaced before they spoil or dry hard in the cup. Hatchlings and juveniles: offer fresh complete gecko diet daily; offer appropriately sized insects about 2-3x/week Adults: offer fresh complete gecko diet every other day or about 3x/week; offer insects about 1-2x/week Useful feeders include dubia or discoid roaches, crickets, black soldier fly larvae, silkworms, and small hornworms. Rotate feeder species when possible and gut-load insects for 24-48 hours before feeding.

Hard-bodied insects should generally be no wider than the space between the gecko's eyes.

Waxworms and other very fatty feeders should be treats rather than staples.

Wild gargoyle geckos sometimes consume small vertebrates, but vertebrate prey is not necessary for a healthy captive diet and should not be used as a routine food. Complete gecko diet plus appropriately supplemented insects is easier to balance and carries less risk of excessive calories.

Baby food and plain fruit should never replace a complete gecko diet; fresh fruit can be an occasional enrichment treat only.

Supplements

A genuinely complete commercial gecko diet should be mixed exactly as directed and usually needs no extra calcium or vitamins added to it. Lightly supplement separately offered insects according to a compatible program. Extra powders can unbalance a complete diet.

Coordinate dietary D3 with measured UVB and the diet’s existing vitamin content. Use a balanced reptile multivitamin only where the insect/diet program calls for it; follow label directions and avoid combining multiple high-dose products.

Decor & Enrichment

Decor is functional habitat, not decoration alone. Use cork bark, sturdy branches, vines, ledges, broad leaves, and plants to create covered routes from the bottom to the top of the enclosure. Include several sleeping retreats and perches at different temperatures and light levels so the gecko can choose where

to rest. Horizontal and diagonal branches are especially important because they provide stable footing for this relatively heavy-bodied species.

Handling & Behavior

Let the animal settle and feed consistently before brief optional handling. Support the body, work low over a safe surface and stop when repeated retreat, escape attempts or defensive behavior show stress. Never lift a lizard by its tail.

Avoid handling immediately after meals, during difficult sheds or when unwell. Wash hands and use dedicated equipment to reduce disease transfer.

Cleaning & Routine Maintenance

Daily: check the animal's access to water/appropriate moisture, temperatures, visible condition and equipment. Remove feces, contaminated material, dead feeders and spoiled food promptly.

Regularly: clean dishes and feeding surfaces, check thermostats and probes, inspect decor stability and maintain ventilation. Bioactive systems still require waste checks, food removal, plant care and drainage maintenance; they are not self-cleaning.

Record appetite, activity, shed history and weight or body condition when safe. Arrange veterinary assessment for unexplained changes. Quarantine new animals with dedicated tools, and disinfect only with a suitable product used at the label dilution/contact time, then rinse and dry as required before return.

Cohabitation

Individual housing is the safest routine default. Shared housing can cause food competition, injury or reproductive pressure even without obvious fighting. Never assume a male/female pair or female group is compatible. Breeding introductions belong in a planned specialist program with separate backup enclosures and individual monitoring.

Shedding & Seasonal Changes

Maintain access to a clean humid retreat and inspect toes, tail tip and eyes after shedding when observation is possible. Do not peel retained skin or apply oils/adhesives. Persistent constriction, damaged tissue or repeated failed sheds warrants veterinary help.

Seasonal changes in appetite or activity can occur. Check temperatures, hydration and weight before treating prolonged refusal as normal. Deliberate brumation or breeding cycles require healthy animals and species-specific preparation; pet care does not require forcing a seasonal shutdown.

Reproduction

Mature females may develop eggs, including infertile eggs without a current male. Provide a suitable moist laying retreat or substrate access, monitor body condition and nutrition, and seek veterinary advice for prolonged straining, swelling or decline. Breeding, incubation and offspring rearing require a prepared species-specific program.

Common Health Concerns

- Poor body condition or feeding decline: check the full gradient, hydration and food quality; arrange assessment for persistent refusal accompanied by weight loss.
- Calcium/vitamin imbalance: weakness, tremors, a soft or misshapen jaw or abnormal limbs require veterinary assessment; extra high-dose supplements are not a substitute for diagnosis.
- Skin, shed or injury problems: repeated retained skin, wounds, swelling or abnormal discoloration deserve prompt attention.
- Respiratory or systemic illness: labored/open-mouth breathing at rest, marked lethargy, loss of balance, persistent diarrhea or rapid deterioration require an experienced exotic veterinarian.

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